



HULT
INTERNATIONAL
BUSINESS SCHOOL

A1.1: Designing a Database for your Client.

(Technical Document)

Business Challenge I - Data Analysis Challenge

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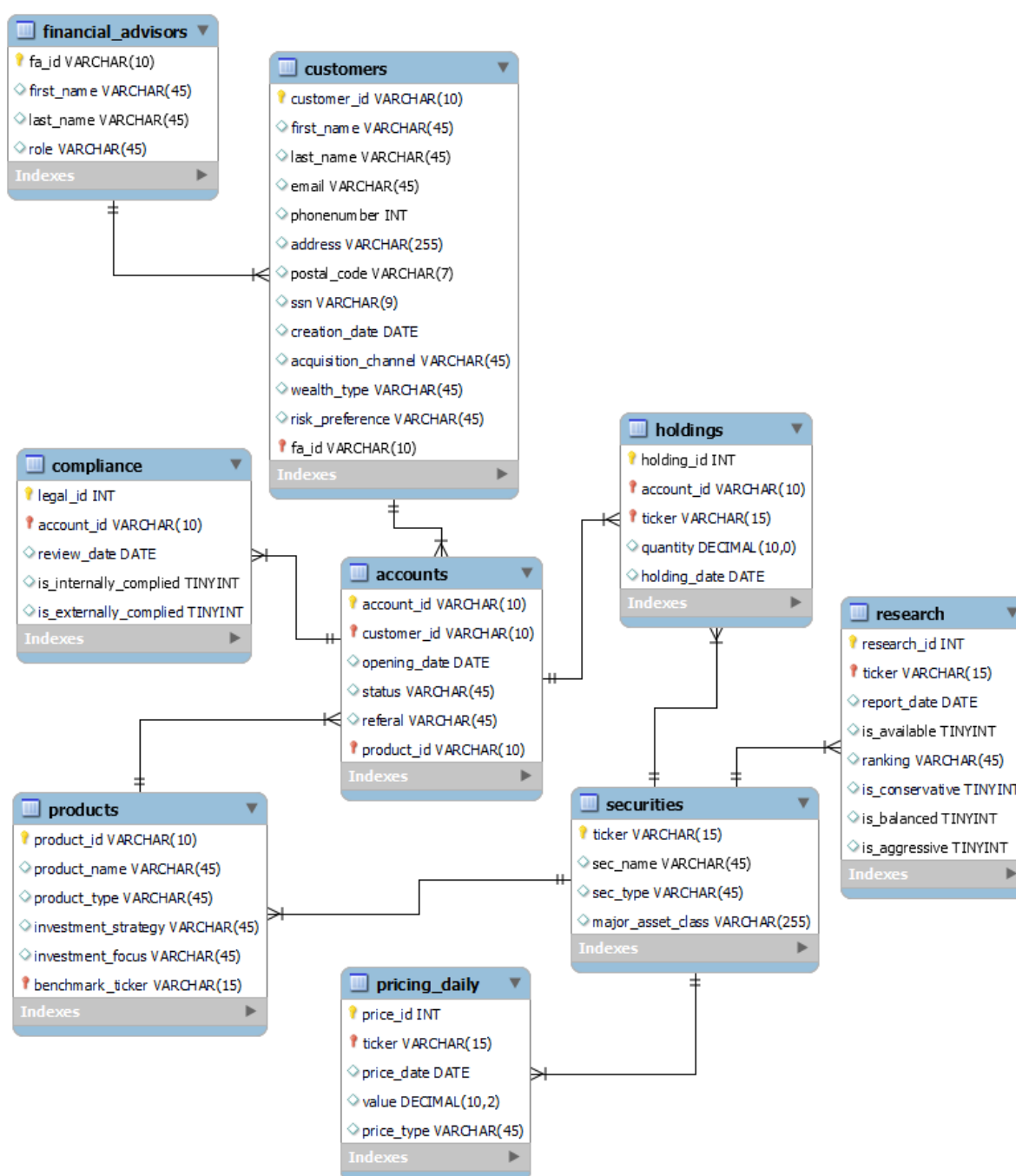
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The Entity Relationship Diagram (ERD)

The ERD comprises ten interconnected tables linked via Primary Keys (PK) and Foreign Keys (FK), allowing data to flow seamlessly between them. This structure is intended to support critical services such as customer management, financial advisory, product research, investment monitoring, and compliance, all of which are necessary for smooth operations and informed decision-making.



1. Table: Customers

Field Name	Function
customer_id	Uniquely identifies each customer.
first_name	Stores customer's first name for personalization.
last_name	Completes the customer's name for formal identification.
Email	Contact point for communication and updates.
Address	Customer residence
Postal_code	Residence area postal code
Ssn	Social security number of the customer
Creation_date	Customer registration date
acquisition_channel	Medium of acquisition of the customer (walkin, sales call, marketing ad etc)
phonenumber	Records the customer's phone number for communication.
wealth_type	Categorizes customers based on wealth tiers: Affluent, High Net Worth, or Ultra-High Net Worth.
risk_preference	Captures the customer's investment risk appetite and ability: Conservative, Balanced, or Aggressive.
fa_id	Links to the financial_advisors table to assign an advisor.

The customers table is the foundation for managing customer-specific data and serves as Wells Fargo's central hub for operations.

The customer_id uniquely identifies everyone, anchoring customer data throughout the system and allowing for precise tracking.

The table categorizes customers based on their risk preference profile and wealth type, ensuring that they are aligned with appropriate financial products that meet their needs.

Furthermore, the fa_id field connects each customer to a financial advisor, allowing for personalized relationship management and increased customer satisfaction through dedicated advisory support.

Data Flow

- Customers are added and recorded in the customer table.
- Financial advisors are assigned based on fa_id from the financial_advisors table.

2. Table: Financial Advisors

Field Name	Function
fa_id	Uniquely identifies each financial advisor.
first_name	Captures the advisor's first name.
last_name	Completes the advisor's full name for identification.
role	Records the position of the advisors in the bank

The financial_advisors table is critical for managing financial advisor information, and it serves as the foundation for Wells Fargo's advisor-customer relationships.

The fa_id serves as a unique identifier for all advisor-related records, ensuring consistent tracking and integration across customer relationships.

This system provides personalized and effective advisory services tailored to each customer's financial goals by connecting advisors to customers via the customers table.

The role field differentiates between advisor levels, allowing for structured workload distribution and the assignment of experienced advisors to high-value customers.

Data Flow

- The financial_advisors table contains advisor information. The fa_id field connects advisers and customers in the customers table, resulting in structured relationships.
- The role field helps maintain organizational hierarchy by assigning advisors depending on their experience and skills.

Data Flow Path: financial_advisors → customers → Customer Relationships & Portfolio Monitoring

3. Table: Accounts

Field Name	Function
account_id	Uniquely identifies each account.
customer_id	Links the account to the owner in the customers table.
opening_date	Tracks the account creation date for reporting.
status	Captures the account's lifecycle stage: Active, Closed, or Suspended.
referral	How did the customer get to know about this product: FA advisor, website, marketing campaign
product_id	Links the account to the products table for product tracking.

The accounts table is an important link between customers, financial products, and investments. It makes it easy to manage account lifecycle statuses (like "Active" or "Closed"), keep an eye on portfolios by linking accounts to products and holdings, and keep track of compliance to make sure that rules are followed.

The account_id is like a central anchor that makes sure all the data in the system flows in the same way. The referral field makes it easier to track goal-based acquisitions, which makes marketing strategies more effective.

Data Flow

- When a customer opens an account, their information is stored in the accounts table.
- The account is linked to the customer via the customer_id foreign key.
- The account is linked to a financial product via product_id.
- The holdings table tracks this account's holdings, while the compliance table documents any compliance issues that arise.

4. Table: Holdings

Field Name	Function
holding_id	Uniquely identifies each holding record.
account_id	Links to the owning account.
ticker	Connects to the security in the securities table.
quantity	Tracks the number of securities held.
holding_date	records the date the holding was acquired.

Wells Fargo's holdings table is a key component of portfolio management, connecting customer accounts to specific securities and tracking critical investment details such as quantity and acquisition dates (holding_date).

By integrating with the securities and pricing_daily tables, it allows for accurate portfolio valuation. The account_id field ensures that each holding is correctly associated with the appropriate customer portfolio, while the ticker field links holdings to securities for seamless pricing integration.

The quantity field contains information for calculating investment value and assessing exposure, while the holding_date field allows advisors to evaluate performance trends and optimize trading strategies.

Data Flow

- When a customer purchases a security, the details are recorded in the holdings table, linked to the account via account_id and the security via ticker.
- Using pricing data from the pricing_daily table and the ticker field, the holdings value is calculated.

Data Flow Path: accounts → holdings → securities → pricing_daily → Portfolio Valuation

5. Table: Securities

Field Name	Function
ticker	Unique identifier for each security.
security_name	Descriptive name of the security (e.g., Apple Inc.).
security_type	Type of security (e.g., Equity, ETF).
major_asset_class	High-level categorization of securities (Health, Tech, etc.).

The securities table lists all kinds of financial instruments, like stocks, bonds, and mutual funds.

It is very important for portfolio analysis because it connects securities to holdings, makes it easier to track performance, and helps divide the market by putting securities into asset classes that make portfolio structuring better.

The ticker is a unique number that makes sure accurate links to holdings and creates a central security catalog. The major_asset_class field also helps to diversify portfolios, which lets clients use structured and strategic investment strategies that are specific to their goals.

Data Flow

- Securities are kept in the securities table.
- In the holdings table, customers can refer to their securities by tickers.
- Analyze performance and pricing data using related pricing tables.

Data Flow Path: securities → holdings → pricing_daily → research → Advisors' Insights

6. Table: Products

Field Name	Function
product_id	Uniquely identifies each financial product.
product_name	Describes the product (e.g., Retirement Plan).
product_type	Categorizes the product (e.g., Tax Saving, High Growth).
Investment_strategy	Defines the strategy associated with the product (e.g., Conservative, Balanced, Aggressive).
Investment_focus	Indicates the sector or area the product focuses on (e.g., Equity, Fixed Income or theme-focused like ESG).
benchmark_ticker	Links to the securities table to track the benchmark for performance evaluation.

The products table catalogs all of Wells Fargo's financial products, including retirement funds, savings plans, and tax-saving instruments, and serves as a centralized repository for recommendation and performance analysis.

The product_type field aligns offerings with customer acquisition goals, such as retirement planning or wealth creation, allowing advisors to make purpose-driven recommendations based on individual financial goals.

Additionally, the benchmark_ticker field connects products to market benchmarks, allowing for real-time performance comparisons.

The investment_focus field improves sector-specific analysis based on investment_strategy, allowing advisors to match products to customer preferences or emerging market trends.

Data Flow

- Financial products are categorized in the products table.
- Accounts use product_id to align and track goals.
- Benchmarks are linked through the benchmark_ticker field, which references the securities table to evaluate performance.
- Advisors use the investment_strategy and investment_focus fields to match customers' goals and preferences with relevant products.

Data Flow Path: products → accounts → securities → Benchmark Comparisons

7. Table: Research

Field Name	Function
research_id	Uniquely identifies each research record.
ticker	Links to the securities table identifying the security being analyzed.
report_id	Identifier for the specific research report.
is_available	Indicates whether the security is currently available to customers (1: Yes, 0: No).
ranking	Assigns a performance rank to the security based on research findings (Buy, Sell, Hold)
is_conservative	Flags if the security aligns with conservative strategies (1: Yes, 0: No).
is_balanced	Flags if the security aligns with balanced strategies (1: Yes, 0: No).
is_aggressive	Flags if the security aligns with aggressive strategies (1: Yes, 0: No).

The research table serves as a single location for verified insights, analysis, and reports on specific securities, enhancing Wells Fargo's financial advisory services with data-driven investment strategies.

By connecting research insights to securities via the ticker field, the table ensures advisors have accurate data to make informed buy or sell decisions.

The is_available field verifies the readiness of securities for trade, ensuring that background checks and analysis are completed to avoid issues such as companies withdrawing their stocks.

Furthermore, fields such as is_conservative, is_balanced, and is_aggressive categorize securities based on risk strategies, allowing advisors to tailor investment recommendations to customer risk profiles.

Data Flow

- Research reports are created, verified, and saved in the research table.
- The ticker field links research data with securities in the securities table.
- Advisors retrieve verified insights (is_available) to ensure securities are trade-ready before making recommendations.
- The risk categorization fields (is_conservative, is_balanced, is_aggressive) align with customer preferences in the customers table, allowing for tailored recommendations.

Data Flow Path: securities → research → Advisors' Recommendations → Customer Portfolios

8. Table: Pricing Daily

Field Name	Function
pricing_id	Uniquely identifies each price record.
ticker	Links to the securities table to identify the security.
price_date	Records the date of the price update.
value	Captures the value of the security for the day.
price_type	Gives the type of the price value (e.g. adjusted close).

The pricing_daily table is required for storing daily price data for securities, resulting in a historical record of market movements that allows for accurate valuation and trend analysis.

The value field records the daily price of securities, ensuring precise portfolio valuation, whereas the price_date field tracks prices over time, allowing advisors and analysts to identify trends, assess volatility.

Furthermore, the price_type field (e.g., Open, Close, High, Low, Adjusted) provides detailed insights, allowing for advanced analysis to optimize buy/sell recommendations and trading strategies.

Data Flow

- The pricing_daily table records daily price data for securities using the ticker field.
- Pricing data is sent to the security table for security-specific performance tracking.
- Use the holdings table to calculate portfolio valuations based on recent prices.
- Analyze historical pricing trends to improve buy/sell recommendations.

Data Flow Path: securities → pricing_daily → holdings → Portfolio Valuation & Recommendations

9. Table: Compliance

Field Name	Function
legal_id	Uniquely identifies each compliance record.
account_id	Links compliance records to the corresponding account in the accounts table.
review_date	Records the date when compliance was reviewed for the account.
is_internally_complied	Flags if the customer account complies with internal bank rules and policies (1: Complied, 0: Not Complied).
is_externally_complied	Flags if the bank complies with government laws and regulations (1: Complied, 0: Non-Complied).

The compliance table is critical for monitoring legal and regulatory activities related to customer accounts, ensuring that Wells Fargo adheres to industry standards and legal frameworks.

It monitors internal compliance using the `is_internally_complied` field, which confirms that accounts adhere to the bank's operational standards, while the `is_externally_complied` field ensures adherence to external regulations imposed by the government and regulatory bodies.

The `review_date` field facilitates timely audits and reviews, allowing for proactive identification and resolution of noncompliance issues.

By linking compliance data to the accounts table, this system enables early detection of high-risk accounts, thereby improving risk mitigation efforts.

Data Flow

- Compliance records are created and saved in the compliance table.
- The `account_id` identifies compliance data for specific accounts in the accounts table.
- Compliance flags (`is_internally_complied` and `is_externally_complied`) are updated during reviews.
- Compliance review dates (`review_date`) align with audit schedules to ensure regulatory and operational standards are met.

Data Flow Path: accounts → compliance → Internal & External Review Reports

SQL Database Connection Credentials.

Database credentials

hostname = hult-team7.mysql.database.azure.com

port = 3306

username = prof

password = clever_password

database_name = Wellsfargo

Data Source:

1. Table Customers	: The data in this table is a dummy data set.
2. Table Financial Advisors	: The data in this table is a dummy data set.
3. Table Accounts	: The data in this table is a dummy data set.
4. Table Holdings	: The Data is collected from 13F fillings of Point72 Asset Management
5. Table Securities	: The data is collected from yahoo finance through excel functions.
6. Table Products	: The data in this table is a dummy data set.
7. Table Research	: The data in this table is a dummy data set.
8. Table Pricing Daily	: The data is collected from yahoo finance through python code.
9. Table Compliance	: The data in this table is a dummy data set.

References:

1. *Point72 Asset Management, L.P. 13F Filings.*

(n.d.). <https://13f.info/manager/0001603466-point72-asset-management-l-p>